NAME: ____________

PRACTICE FINAL

Please show your work. External assistance (aside from asking me for clarification) is forbidden. Seek the answers within yourself...

0: Please go through all your homeworks, quizzes, your midterm, and the practice midterm, and make sure you understand how to do **all** the problems.

1: Let $A = \begin{bmatrix} -3 & -4 \\ -4 & -3 \end{bmatrix}$.

(a): Please find the eigenvalues and eigenvectors of A.

For 2×2 matrices, it's OK to use the characteristic polynomial if the eigenvalues aren't completely obvious. The trace-formula is a convenient way to get the characteristic polynomial. Applied here, we get
$\mathrm{CharPoly}(\lambda) = \lambda^2 - (-3-3)\lambda + ((-3)(-3) - (-4)(-4)) = \lambda^2 + 6\lambda - 7 = (\lambda-1)(\lambda+7)$.
So the eigenvalues are clearly $\boxed{\{1, -7\}}$. To get the corresponding eigenvectors, we need to find the right nullspaces of the matrices

$$\begin{bmatrix} -3-1 & -4 \\ -4 & -3-1 \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

and

$$\begin{bmatrix} -3-(-7) & -4 \\ -4 & -3-(-7) \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}.$$

Skipping the details (which boil down either to solving by inspection, or computing the reduced row echelon form), it is easy to see that these nullspaces are respectively generated by $\boxed{\begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$.

(b): Let $u_{k+1} := A^{-1}u_k$. What is the limit of u_k as $k \longrightarrow +\infty$ given the initial condition $u_0 := \begin{bmatrix} 0 \\ 3 \end{bmatrix}$?

The key trick here is to use diagonalization to easily compute the powers we need to evaluate. In particular, since $V^{-1}AV = \Lambda$ when V is the matrix whose columns are the eigenvectors of A and Λ is the diagonal matrix consisting of the eigenvalues (in order corresponding to the columns of V), we easily obtain $A = V\Lambda V^{-1}$. (Note that the identity $V^{-1}AV = \Lambda$ only applies when there are n linearly independent eigenvectors and A is $n\times n$!) For our matrix at hand, we thus obtain

$$\begin{bmatrix} -3 & -4 \\ -4 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}/2.$$

Taking arbitrary powers is easy now: $(A^{-1})^k = V\Lambda^{-k}V^{-1}$ so we obtain

$$\begin{bmatrix} -3 & -4 \\ -4 & -3 \end{bmatrix}^{-k} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix}^{-k}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}/2 = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{(-7)^k} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}/2.$$

So the limit is $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\lim\limits_{k\longrightarrow\infty}\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{(-7)^k} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\frac{1}{2}\begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 3/2 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 3/2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} -3/2 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} -3/2 \\ 3/2 \end{bmatrix}}.$$

2: Please find a basis for the subspace of $\mathbb{R}^4$ perpendicular to the span of $\begin{bmatrix}1\\1\\1\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$.

The key here is that two vectors are perpendicular (in $\mathbb{R}^4$) if and only if their dot product is 0. So then, any column vector $x \in \mathbb{R}^4$ perpendicular to both the stated vectors must satisfy $\begin{bmatrix}1 & 1 & 1 & 3\\1 & 1 & 1 & 1\end{bmatrix} x = \begin{bmatrix}0\\0\end{bmatrix}$. In other words, we just have to find a basis for the right nullspace of $\begin{bmatrix}1 & 1 & 1 & 3\\1 & 1 & 1 & 1\end{bmatrix}$. Via a little Gaussian Elimination, we get a row-echelon form of $\begin{bmatrix}1 & 1 & 1 & 3\\0 & 0 & 0 & 1\end{bmatrix}$, and then we get a reduced row-echelon form of $\begin{bmatrix}1 & 1 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}$ (which has free columns in positions 2 and 3). So we easily obtain a basis of $\boxed{\left\{\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\\0\end{bmatrix}\right\}}$.

3: The following information is known about a matrix A:

$$A\begin{bmatrix}1\\2\\3\\4\end{bmatrix}=\begin{bmatrix}-1\\1\end{bmatrix} \text{ and } A\begin{bmatrix}2\\3\\4\\1\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}.$$

(a) What is the dimension of the right-nullspace of A? Please explain why.
(b) What is the dimension of the row space of A? Please explain why.
(c) What is the dimension of the left-nullspace of A? Please explain why.

(a), (b), and (c) will ultimately follow easily from the Four Subspaces Theorem , or the Rank-Nullity Theorem, depending on one's tastes. But before applying such theorems, we need to see if we can get any information at all about any of the four subspaces. One thing we can derive easily is that A has to be a 2×4 matrix, since only a 2×4 matrix can be multiplied on the right by a 4×1 column vector to yield a 2×1 column vector. We can then easily derive that the column space of A has dimension 2: $\dim \mathrm{ColSpace}(A)\leq 2$ since $\mathrm{ColSpace}(A)\subseteq\mathbb{R}^2$, and $\dim \mathrm{ColSpace}(A)\geq 2$ because $\mathrm{ColSpace}(A)$ contains two linearly independent vectors. The latter fact is obvious since $\begin{bmatrix}-1\\1\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are clearly not multiples of each other.

Now, since the Four Subspaces Theorem tells us that $\dim \mathrm{RowSpace}(A)=\dim \mathrm{ColSpace}(A)$ and $\dim \mathrm{RowSpace}(A)+\dim \mathrm{RightNull}(A)=(\#$ of columns of $A)$, and $2+2=4$, we see that $\boxed{\dim \mathrm{RightNull}(A)=2}$ and $\boxed{\dim \mathrm{RowSpace}(A)=2}$.

The Four Subspaces Theorem also tells us that $\dim \mathrm{ColSpace}(A)+\dim \mathrm{LeftNull}(A)=(\#$ of rows of $A)$. So, since $2+0=2$, we obtain $\boxed{\dim \texttt{LeftNull}(A)=0}$.

(d) Please give a solution, x, to $Ax=\begin{bmatrix}0\\2\end{bmatrix}$.

$$\begin{bmatrix}0\\2\end{bmatrix}=\begin{bmatrix}-1\\1\end{bmatrix}+\begin{bmatrix}-1\\1\end{bmatrix}=A\begin{bmatrix}1\\2\\3\\4\end{bmatrix}+A\begin{bmatrix}2\\3\\4\\1\end{bmatrix}=A\left(\begin{bmatrix}1\\2\\3\\4\end{bmatrix}+\begin{bmatrix}2\\3\\4\\1\end{bmatrix}\right)=A\begin{bmatrix}3\\5\\7\\5\end{bmatrix}.$$

So $\boxed{\begin{bmatrix}3\\5\\7\\5\end{bmatrix}}$ is an answer.

4:

(a) Please evaluate $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

(b) Please derive a formula for $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

(c) What is the coefficient of $x_2 x_4$ in $\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} a & r & s & t \\ r & b & u & v \\ s & u & c & w \\ t & v & w & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$?

(d) Suppose $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix and (λ_1, v_1) are (λ_2, v_2) are eigenpairs with $\lambda_1 \neq \lambda_2$. Please prove that v_1 and v_2 are orthogonal. **Hint:** Try comparing $v_1^\top A v_2$ and $(Av_1)^\top v_2$, and evaluating them in more than way.

(e) Suppose $v_1, \ldots, v_n \in \mathbb{R}^n$ form an orthonormal basis and $V = [v_1, \ldots, v_n]$. Please evaluate $V^\top V$ and $VV^\top$.

(f) Suppose $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix with eigenvectors $v_1, \ldots, v_n \in \mathbb{R}^n$ *forming an orthonormal basis*, $V = [v_1, \ldots, v_n]$, and $x = \begin{bmatrix} x_1 \\ \vdots \\ v_n \end{bmatrix}$. Please evaluate $(Vx)^\top A (Vx)$.

Recall that the singular values of a matrix $A \in \mathbb{R}^{n\times n}$ are the *square roots* of the eigvenvalues of $A^\top A$.

5: Please clearly sketch the image of the unit circle under the linear map $L : \mathbb{R}^{2\times 1} \longrightarrow \mathbb{R}^{2\times 1}$ defined by $L(x) := \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix} x$. In particular, to receive full credit, you must find the...

(a) singular values of the matrix $A := \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$

(b) vectors forming the principal axes of the ellipse defined by $\left\{ L\left(\begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \right) \right\}_{\theta\in[0,2\pi)}$

(c) the maximum and minimum of $|Ax|$ as x ranges over the unit circle.

This problem reduces to finding the maximum and minimum value of $|Ax|^2$ as x ranges over the unit circle in $\mathbb{R}^2$, and finding where this occurs. In the end, this reduces to finding the eigenvalues and eigenvectors of $A^T A$: The maximum (resp. minimum) of $|Ax|$ is the nonnegative square roots of the largest (resp. smallest) eigenvalue of $A^T A$, and this value is attained at a suitable multiple of the corresponding eigenvector of $A^T A$.

For the matrix at hand, we obtain $A^T A = \begin{bmatrix} 18 & 30 \\ 30 & 68 \end{bmatrix}$. The trace formula then tells us that the characteristic polynomial is $\lambda^2 - (18+68)\lambda + (18\cdot 68 - 30^2) = \lambda^2 - 86\lambda + 324$. So the eigenvalues are $\frac{86\pm\sqrt{86^2-4\cdot 324}}{2} = 43 \pm 5\sqrt{61}$. To find the eigenvectors, let's start with the eigenvalue $43+5\sqrt{61}$ and compute the right nullspace of

$$A - \lambda I = \begin{bmatrix} 18-(43+5\sqrt{61}) & 30 \\ 30 & 68-(43+5\sqrt{61}) \end{bmatrix} = \begin{bmatrix} -25-5\sqrt{61} & 30 \\ 30 & 25-5\sqrt{61} \end{bmatrix}.$$

Clearly, $(-25-5\sqrt{61})30 + 30(25+5\sqrt{61}) = 0$. Furthermore,

$$30^2 + (25-5\sqrt{61})(25+5\sqrt{61}) = 900 + 25^2 - (5\sqrt{61})^2 = 900 + 625 - 25\cdot 61 = 0.$$

So we have $\begin{bmatrix} -25-5\sqrt{61} & 30 \\ 30 & 25-5\sqrt{61} \end{bmatrix} \begin{bmatrix} 30 \\ 25+5\sqrt{61} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as desired. (This is a standard trick: To find a vector perpendicular to $[a,b]$ one can simply use $[b,-a]$. Also, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ having rank 1 automatically implies that $[c,d]\begin{bmatrix} b \\ -a \end{bmatrix} = 0$ too.

Now, since $A^T A$ is symmetric, the other eigenvector of $A^T A$ must be orthogonal to $\begin{bmatrix} 30 \\ 25+5\sqrt{61} \end{bmatrix}$. So $\begin{bmatrix} 25+5\sqrt{61} \\ -30 \end{bmatrix}$ clearly works.

To conclude, we thus see that the image of the unit circle under L is the ellipse with $\boxed{\text{principal axes } \begin{bmatrix} 30 \\ 25+5\sqrt{61} \end{bmatrix} \text{ and } \begin{bmatrix} 25+5\sqrt{61} \\ -30 \end{bmatrix}}$, and the respective scalings are $\boxed{\sqrt{25+5\sqrt{61}} \text{ and } \sqrt{25-5\sqrt{61}}}$.

6: Suppose the following information is known about a matrix A:

$A\begin{bmatrix}1\\2\\4\end{bmatrix} = 9\begin{bmatrix}1\\1\\1\end{bmatrix}$, $A\begin{bmatrix}1\\-3\\9\end{bmatrix} = 2\begin{bmatrix}1\\-3\\9\end{bmatrix}$, and $A\begin{bmatrix}1\\1\\1\end{bmatrix} = 3\begin{bmatrix}1\\1\\1\end{bmatrix}$.

(a): Please find the eigenvalues of A.

(b): Please find the eigenvectors of A.

(c): Is A invertible? Please explain...

(d): Is A diagonalizable? Please explain...

(a) and (b): From what is given, and the definition of eigenvalue and eigenvector, there are 2 obvious eigenvectors: $\begin{bmatrix}1\\-3\\9\end{bmatrix}$ and $\begin{bmatrix}1\\1\\1\end{bmatrix}$, with corresponding eigenvalues 2 and 3. A smart idea then (*instead* of trying to solve for the matrix A) is to check if there are any simple linear combinations that yield a (nonzero) right nullvector. Such a linear combination is actually pretty obvious: The first and last given equations have right-hand sides $9\begin{bmatrix}1\\1\\1\end{bmatrix}$ and $3\begin{bmatrix}1\\1\\1\end{bmatrix}$. So then, it is clear that $A\left(\begin{bmatrix}1\\2\\4\end{bmatrix} - 3\begin{bmatrix}1\\1\\1\end{bmatrix}\right) = \begin{bmatrix}0\\0\\0\end{bmatrix}$. In other words, $\begin{bmatrix}-2\\-1\\1\end{bmatrix}$ is a right-null vector. This is then the same thing as saying that $\begin{bmatrix}-2\\-1\\1\end{bmatrix}$ is an eigenvector with corresponding eigenvalue 0. So we have found 3 distinct values with 3 corresponding (linearly independent) eigenvectors. This answers (a) and (b).

(c): An invertible matrix $n \times n$ A always results in the equation $Ax = \emptyset$ having a ***unique*** solution $x \in \mathbb{R}^n$. So our A is **non**-invertible because it has a nonzero right null vector.

(d): A is 3×3 and has 3 linearly independent eigenvectors. (We know the 3 eigenvectors we found are linearly independent since they correspond to 3 distinct eigenvalues.) So A is diagonalizable. (In fact, A diagonalizes to $\begin{bmatrix}2&0&0\\0&3&0\\0&0&0\end{bmatrix}$.)

We never solved for the matrix A. Doing so would have been unneeded extra work...

7:

Find an orthonormal basis for the subspace of $\mathbb{R}^5$ generated by the following pair of vectors:
$\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$

Here, we just apply Gram-Schmidt in a straightforward way. However, it's a good idea to let a_1 be the vector with *simpler* norm. In particular, the first vector has norm 4, while the second has norm $\sqrt{7}$. So let's set $a_1 := \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix}$. Proceeding as usual, we then set $q_1 := \frac{a_1}{|a_1|} = \begin{bmatrix} 1/4 \\ 3/4 \\ -1/4 \\ 1/2 \\ 1/4 \end{bmatrix}$. The second vector in our basis will then simply be

$$q_2 := \frac{a_2 - (q_1 \cdot a_2)q_1}{|a_2 - (q_1 \cdot a_2)q_1|} = \frac{a_2 - (1+6-1+1)q_1/4}{|a_2 - (1+6-1+1)q_1/4|} = \frac{a_2 - \frac{7}{4}q_1}{|a_2 - \frac{7}{4}q_1|} = \begin{bmatrix} 1-7/16 \\ 2-21/16 \\ 1+7/16 \\ 0-7/8 \\ 1-7/16 \end{bmatrix} \Big/ |a_2-(1+6-1+1)q_1/4|$$

$$= \begin{bmatrix} 9/16 \\ 11/16 \\ 23/16 \\ -14/16 \\ 9/16 \end{bmatrix} \Big/ \left(\sqrt{9^2+11^2+23^2+14^2+9^2}/16\right) = \begin{bmatrix} 9 \\ 11 \\ 23 \\ -14 \\ 9 \end{bmatrix} \Big/ \sqrt{1008} = \begin{bmatrix} 9 \\ 11 \\ 23 \\ -14 \\ 9 \end{bmatrix} \Big/ \left(12\sqrt{7}\right)$$

So our orthonormal basis is then simply $\boxed{\left\{ \begin{bmatrix} 1/4 \\ 3/4 \\ -1/4 \\ 1/2 \\ 1/4 \end{bmatrix}, \begin{bmatrix} 3/(4\sqrt{7}) \\ 11/(12\sqrt{7}) \\ 23/(12\sqrt{7}) \\ -\sqrt{7}/6 \\ 3/(4\sqrt{7}) \end{bmatrix} \right\}}.$

8:

Suppose $A \in \mathbb{R}^{m \times n}$ has rank m, $m > n$, $x \in \mathbb{R}^{n \times 1}$, $b \in \mathbb{R}^{m \times 1}$.